

ETHAN WONG

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AI and data professional with a UT Austin analytics master's background, experienced in building data pipelines and LLM-enabled workflows that combine automation with human oversight. Supports enterprise decision-making with forecasting/valuation analytics and builds systems that tie together retrieval, orchestration, and QA processes. Strong Python/SQL foundation, high attention to detail, and a bias toward measurable quality and repeatable workflows.

WORK EXPERIENCE

Analyst, Portfolio Management – Amherst

August 2025 – Present

- Maintained SQL/Databricks pipelines tracking 1,300+ homes actively being prepared and listed for sale across 16 investment funds, standardizing data infrastructure to support a company-wide CRM migration
- Authored SQL scripts querying internal databases to surface 10 pricing metrics for all homes with an active or historical retail sale designation, refreshed weekly to support pricing decisions by retail strategy and investment teams
- Developed a SQL-driven monthly model generating projected retail-sale dates and closing probabilities for 47,000+ single-family homes across the full Amherst portfolio, directly informing retail sales forecasts and budgets
- Automated quarterly fair-market valuations for 47,000+ single-family homes by consolidating multiple Excel models into a single Python pipeline, reducing model turnaround time from 10 hours to 20 minutes
- Managed offshore team execution on a monthly portfolio operations report delivered to the executive leadership team, reviewing figures for accuracy and ensuring revenue, expense, and resident performance metrics tied out across the deck
- Instantiated AI-powered automation to consolidate investor approval packages across 4 distinct deal stages in the retail sale pipeline, cutting manual processing time from 1 hour to under 5 minutes per package while eliminating human error
- Supported quarterly production of capital structure and investment return reporting distributed to the executive leadership team, building and maintaining the underlying Excel models across multiple fund structures
- Designed a 7-field classification system in Salesforce to capture and standardize the reason a home is removed from the retail sale pipeline, enabling historical trend analysis that was previously not possible due to absent data coverage
- Partner with cross-functional teams and global counterparts (India, Costa Rica) to execute monthly/quarterly reporting and respond to ad hoc investor requests

Intern, Portfolio Management and Modeling – Amherst

June 2025 – August 2025

- Quantified and presented YoY property and flood-insurance growth trends for 44,000+ homes by state, market, and fund
- Benchmarked LLMs on 22,000+ resident survey records and deployed the best model via SQL stored procedures into a firm-wide Tableau dashboard, reducing misclassification by 20% across multiple targets

Advanced Analytics Programming Grader – University of Texas at Austin

January 2024 – May 2024

- Designed homework and project rubrics to assess over 80 students' mastery of Python libraries (NumPy, Pandas, Matplotlib, Scikit-learn, Statsmodels)

Predictive Analytics and Data Mining Teaching Assistant – University of Texas at Austin

August 2023 – May 2024

- Refined and taught the professor's Scikit-learn class material and homework to promote clarity and streamline course delivery
- Constructed and led class quiz review sessions as a resource for over 250 students across 8 different course offerings

RELEVANT PROJECT EXPERIENCE

Forecasting Renter Retention for Amherst

January 2025 – April 2025

- Created predictive models (CatBoost, XGBoost, FNN) to forecast renter retention and turnover costs, optimizing expenses and stabilizing cash flow
- Leveraged lease and resident data with external sources to uncover drivers of retention and guide decision-making

MemoraVault

March 2025 – April 2025

- Developed a retrieval-augmented study assistant that standardized PDFs, class notes, and YouTube content into a common JSON format, embedded the corpus, and generated grounded responses using indexed course materials
- Implemented vector-based retrieval with FAISS to improve relevance versus traditional keyword search

- Added multimodal extraction for video sources by combining ViT/BLIP with OCR to capture visual/text cues and enrich retrieval metadata

Procurement ERP With AI Integration

March 2025 – April 2025

- Architected an end-to-end design linking SAP (OData) to a Snowflake ETL layer and downstream market forecasting models; prototyped an agent-style LLM workflow to apply business rules and generate procurement recommendations
- Designed a human-review loop where dashboards surface recommendations with confidence signals, driver explanations, and structured override fields that capture manager feedback for iteration
- Established governance and monitoring patterns (data checks, hallucination safeguards, and audit logs for final decisions)

Connect 4 AI Agent Development

January 2025 – April 2025

- Trained CNN and transformer evaluators on 2.4M board states to learn strategy signals, then used model-driven opponents in self-play to boost a reinforcement agent's win rate by 15% against a benchmark.
- Shipped a playable web experience using an Anvil front end and Dockerized inference services deployed on AWS for real-time interaction

Decomposing Experiential Value From Used Cars for Marketing

October 2024 – December 2024

- Collected 15K+ consumer reviews across 175+ vehicle profiles and built an NLP workflow (cleaning, topic discovery, sentiment scoring) to quantify “experiential” value drivers for pricing and positioning
- Upgraded sentiment labeling from NLTK to an LLM-based classifier with structured outputs, entity/attribute extraction, and rating–text consistency checks, reducing validation misclassification by 15%.

Predicting Problematic Internet Use Severity in Children

October 2024 – December 2024

- Built a modular scikit-learn pipeline to streamline experimentation for early screening, automating feature integration from actigraphy data, missing-data handling, and imbalance-aware training
- Implemented objective tuning to emphasize detection of moderate/severe cases, improving recall for higher-priority outcomes

Identifying Customer Churn for NetApp

January 2024 – May 2024

- Built predictive models (ARIMA, LGBM, Poisson) to identify customers likely to reduce spending for targeted outreach
- Presented key insights on declining customer behavior to both technical and non-technical stakeholders weekly

LEADERSHIP AND INVOLVEMENT

Business Analytics Association – Founding Logistics Director, Austin, Texas

August 2022 – May 2024

- Oversaw 100+ member application process, biweekly announcements, room reservations, and inflow data collection
- Organized semesterly Power BI Desktop workshops to enhance members' data manipulation and visualization skills

EDUCATION

The University of Texas at Austin – McCombs School of Business

May 2025

Master of Science, Business Analytics | GPA: 3.95

- Coursework Includes: Time Series Analytics, Advanced Machine Learning, Analytics for Unstructured Data, Information Management, Marketing Analytics, Privacy Preserving Analytics, Optimization, Deep Learning, Unsupervised Learning

The University of Texas at Austin – McCombs School of Business

May 2024

Bachelor of Business Analytics, Minor in Supply Chain Management | GPA: 3.88 (University Honors)

TECHNICAL SKILLS

Computer Software: Excel (Solver, XLSTAT, Macabacus), Power BI, Tableau, Databricks, Salesforce, Snowflake, GitHub

Computer Languages: R, Python (Pandas, Scikit-learn, Gurobi, Sci-Py, TensorFlow/Keras, PyTorch, NLTK, spaCy, VADER), SQL

ADDITIONAL INFORMATION

Interests: Video games, Golf, Swimming, Rowing, Cars, Traveling, Magic the Gathering, Recreational Shooting, Double Bass

Work Eligibility: Eligible to work in the United States with no restrictions